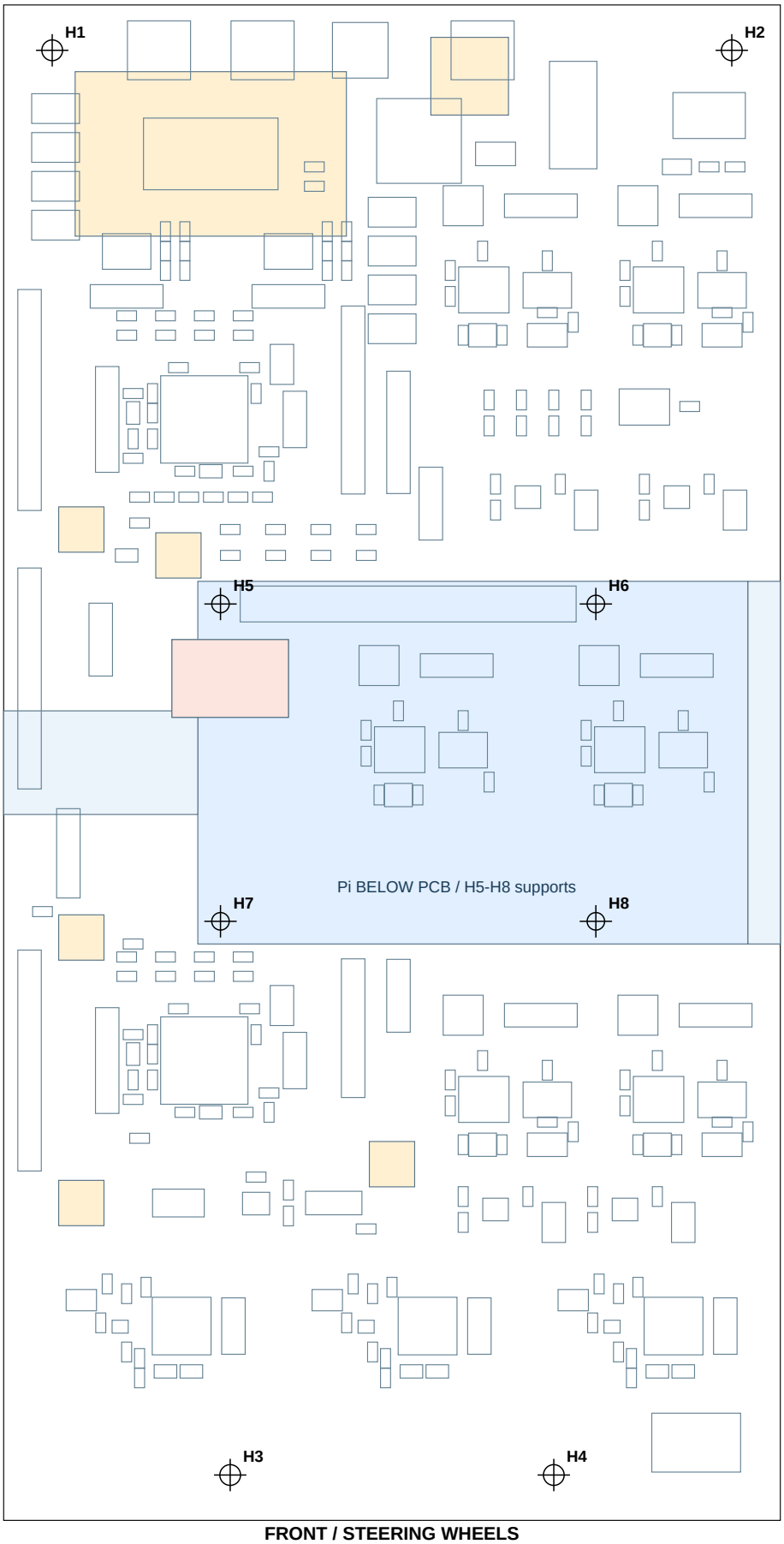
Lab 4 - provisional PCB fit template

Print at 100% / Actual size. Check the 50 mm bar before comparing with the car.

MEASURED HOLE CENTERS - UNROUTED - NOT FOR FABRICATION

REAR / DRIVE MOTORS (0,0) at upper-left; x right, y down



FRONT / STEERING WHEELS

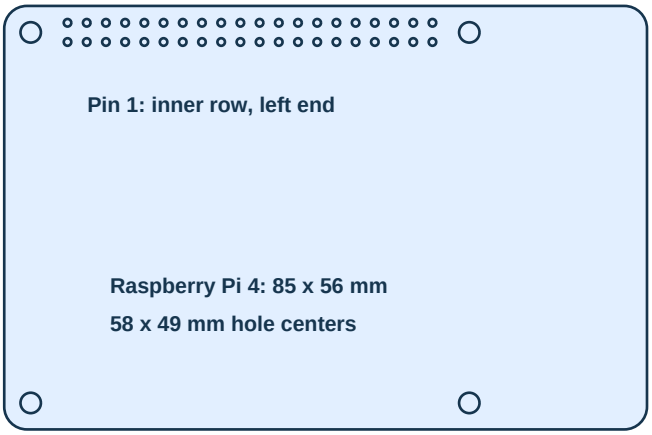
H1-H4: measured centers; M3 PCB drills proposed. Chassis holes measured 4.5 mm.

Check wires, steering travel, Pi below the PCB, port access and separate battery location.

This bar must measure 50 mm.

Pi alignment and vertical fit check

Top view below is 1:1. Place the unpowered Pi component-side-up to compare holes and pins.



USB / Ethernet face right.

GPIO row faces the REAR.

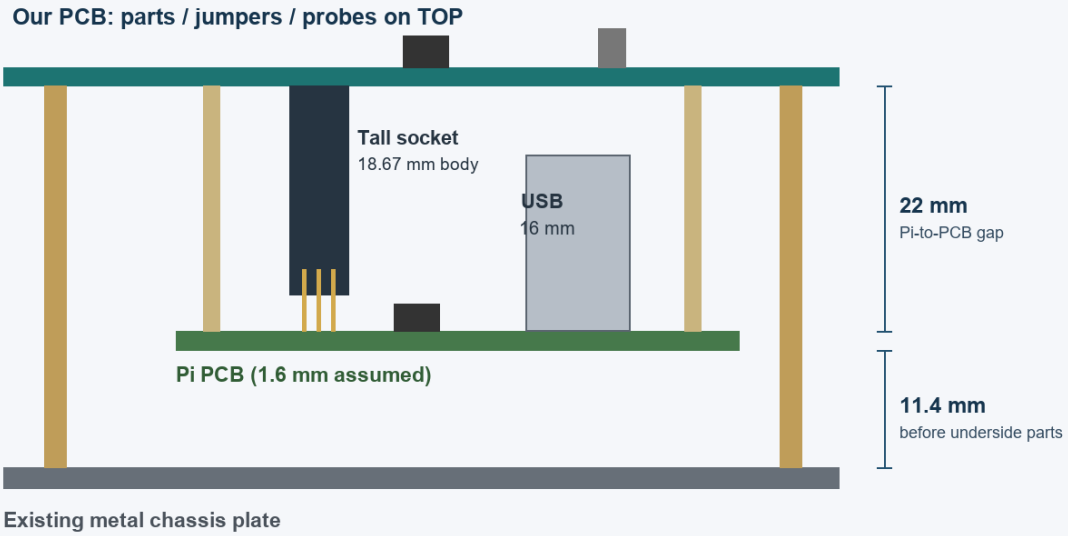
microSD is on Pi underside.

Do not use this as a drill guide.

Main-PCB Pi origin: (30,89) mm.

How the boards stack - proposed dimensions

Pi component side faces UP. J70 socket is soldered to the UNDERSIDE of our PCB.



35 mm chassis standoffs proposed. All heights and mating must be checked on the actual assembly.

Record from the unpowered physical mock-up:

Main-PCB underside above chassis: _____ mm (proposal 35 mm).

Pi top to main-PCB underside: _____ mm (proposal 22 mm).

Lowest Pi part / card / wire above metal: _____ mm.

Tallest Pi feature, including cooler: _____ mm above Pi board.

Compare H1-H4 with measured chassis holes; do not drill from this print.

Can USB, Ethernet, microSD and cables be reached? _____

Pi mounting hardware OD <=5.5 mm proposed. Review the course 3 mm hole-clearance interpretation before fabrication.